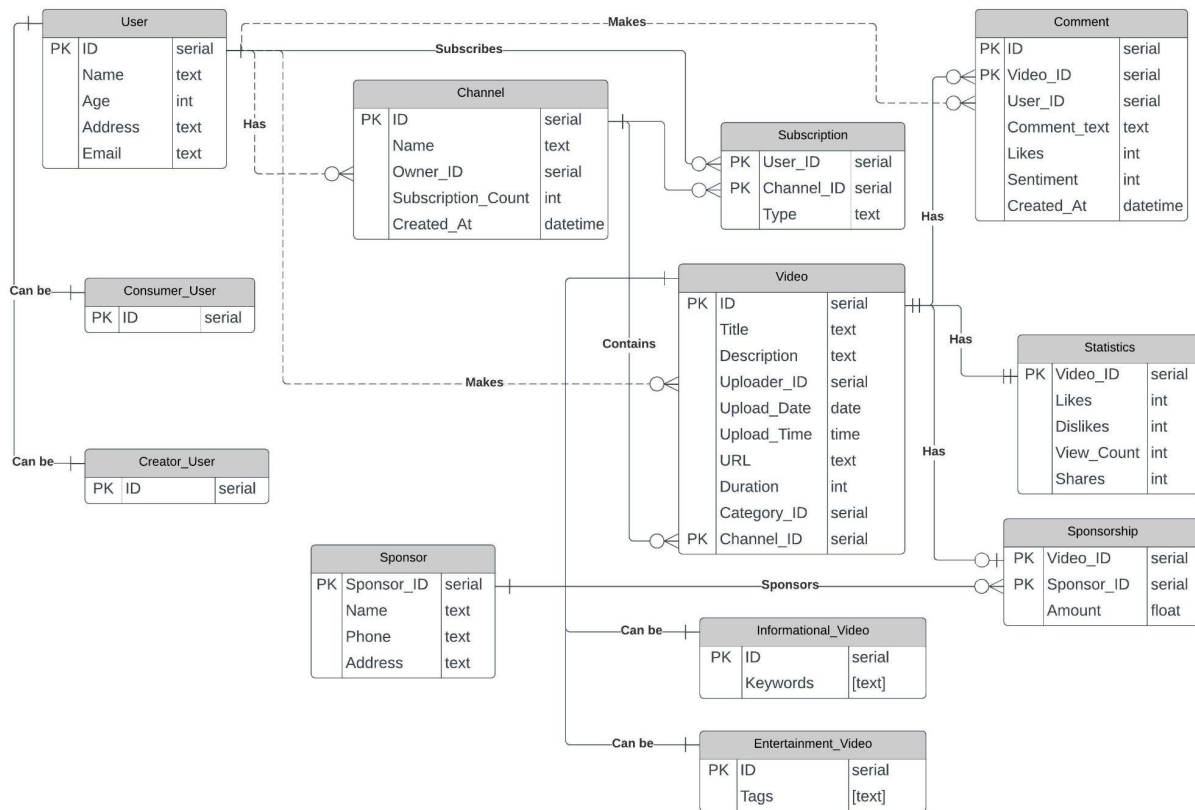


CSCI 585 HW1



Note: Due to the requirement of EERD, subtypes have been included in the Diagram.

Subtypes included are for the User type and Video type, Consumer/Creator and Informational/Entertainment respectively. It is assumed that every user is by default a consumer and optionally a creator due to how YouTube is formatted.

Caveat: Having included subtypes for the user and video types, there wasn't a clear way to indicate the relationship between the super and subtypes without a line that included crow's feet notation. As a result the final decision was to leave the crow's feet notation in for the connection between super and sub types but make sure that the connection is 1:1 for each (1 as in at least 1 not 1 and only 1).

Along with this, every action a User can take is assumed to be default null:

- A user can watch videos without subscribing or commenting
- A user can have no videos or channels
- A video can have zero or one sponsor
- A sponsor can sponsor many videos

A quirk that does not appear in the diagram as detailed below is the existence of bridge tables. This design choice was chosen in an attempt to address the overlapping crow's feet notation symbols that are attached to specific attributes rather than the table itself.

There are indicators of weak/strong relationships which are the dotted and solid lines respectively. It was used to intuitively identify that there can be Comments and Channels and Videos with no Owner/creator entry in the Users table. Though this is more of a design choice on my part, the fact that there exists comments without user accounts on YouTube insinuates that the original user can get deleted or deleted their own account.

Entities:

- User: Represents a unique user on the YouTube platform with attributes such as id, name, email, age, address, etc.
- Channel: Represents a dashboard for a user to upload and distribute videos, with attributes such as name, owner, subscription count, and creation date.
- Video: Represents a video uploaded by a user with attributes such as URL, title, thumbnails, category, duration, description, uploader id, upload date, and upload time.
- Subscription: Represents the relationship between a video consumer and a channel, with attributes such as subscription type (paid or free).
- Category: Represents the type of video (informational or entertainment) with attributes such as keywords or tags.
- Video Statistics: Represents the popularity of a video with attributes such as likes, dislikes, view count, shares, etc.
- Comment: Represents user feedback on a video with attributes such as video id, user id, comment text, likes, sentiment, and comment date.
- Sponsor: Represents a third-party sponsor for a video with attributes such as sponsor id, name, phone, address, and amount sponsored.

ER cardinality

Format: (node 1 - node 2: explanation of cardinality)

- User-Channel: One user can have many channels (1:N).
- Channel-Video: One channel can have many videos (1:N).
- Video-Category: One video can belong to one category (1:1).
- User-Subscription: One user can subscribe to many channels (1:N).
- Video-Statistics: One video can have many statistics (1:N).
- Video-Comment: One video can have many comments (1:N).
- User-Comment: One user can post many comments (1:N).
- Video-Sponsor: One video can have one sponsor (1:1).